



Event-Based Learning and Improvement: Radiology's Move From Peer Review to Peer Learning

Lane F. Donnelly, (MD)*,^{†,*} and Carolina V. Guimaraes, (MD)*

Over the past 15 years, the radiology community has made great progress moving from a system of score-based peer review to one of peer learning. Much has been learned along the way. In peer learning, cases in which learning opportunities are identified are reviewed solely for the purpose of fostering learning and improvement. This article defines peer learning and peer review and emphasizes the difference; looks back at the 20-year history of score-based peer review and transition to peer learning; outlines the problems with score-based peer review and the key elements of peer learning; discusses the current state of peer learning; and outlines future challenges and opportunities.

Semin Ultrasound CT MRI 45: 161–169 © 2024 Elsevier Inc. All rights reserved.

Definitions of Peer Learning and Peer Review

Quality improvement and professional practice evaluation are both key tasks for all health systems.¹ Review of events in which system issues or human error are evaluated is a key component of both quality improvement and the evaluation of clinical providers as pertains to individual provider competence.¹ The sequestration of activities pertaining to event-based evaluation as it pertains to quality improvement from that which pertains to professional practice evaluation is key.^{1–8} Quality improvement relies on transparency and active reporting. Success is dependent on trust in the system.^{1–8} Professional practice evaluation, which questions individual provider competence and potentially puts health care providers' ability to make a living at risk, inherently erodes trust.^{1–4} If one process is created to do both quality improvement and professional practice evaluation that process will inherently be flawed and fail.

Over the past 15–20 years, there has been increased recognition of the need for sequestration of quality improvement

from professional practice evaluation and attention given to creating systems in which the tasks are separated.¹ Related, during that period, the radiology community has moved from a peer review approach to a peer learning approach as it pertains to event review.^{1–8}

For clarity, we will define what is meant by peer review and peer learning, as pertains to this discussion. The broad definition of peer review is “evaluation of professional work by others working in the same field.”^{1–8} Over the past 20 years in radiology, peer review has come to mean the use of score-based evaluation of errors to evaluate individual radiologists' performance.³ Peer learning is defined as the evaluation of radiology cases purely for the purposes of learning and improvement.^{1–8}

To truly understand the nuances between the practice of score-based peer review and peer learning in radiology, we will review:

- The Transition to Peer Learning From Peer Review in the Radiology Community: Looking Back Over the Past 20 Years
- Peer Learning - The Current State
- Future Opportunities and Challenges

*Department of Radiology, University of North Carolina School of Medicine, Chapel Hill, NC.

[†]Department of Pediatrics, University of North Carolina School of Medicine, Chapel Hill, NC.

Address reprint requests to Lane F. Donnelly, MD, University of North Carolina School of Medicine, UNC Health, UNC Children's Hospital, 509 Old Infirmary, Chapel Hill, NC. Email: lane_donnelly@med.unc.edu

The Transition to Peer Learning From Peer Review in the Radiology Community: Looking Back Over the Past 20 Years

The Launch of Score-Based Peer Review

In response to the 2000 *Institute for Medicine* publication *To err is Human*⁹ calling out that error was common in medicine, the American College of Radiology (ACR) convened a workgroup to evaluate potential responses from the radiology community. The workgroup constructed a quality assurance approach in which score-based peer review of previously reported studies was performed to identify and quantify potential errors.^{10,11} That process was termed RADPEER.^{10,11} Using the process, radiologists randomly reviewed their colleagues' cases during the course of routine work.^{10,11} Each case was then given a score from 1-4 based on the level of agreement or disagreement that the reviewer had with the initial reading.^{10,11} A score of 1 meant "concur with interpretation"; 2 meant "difficult diagnosis, not ordinarily expected to be made"; 3 meant "diagnosis should be made most of the time"; and 4 meant "diagnosis should be made almost every time—misinterpretation."^{10,11} Scores of 3 and 4 were considered errors.

RADPEER was piloted in 20 facilities in 2001¹⁰ and steadily spread in use. In a 2009 publication, RADPEER was described as being used by over 10,000 radiologists and more than 800 radiology practices around the country.¹¹

Several "regulatory" changes occurred which catapulted the use of RADPEER or RADPEER-like score-based peer review systems throughout the radiology community. In 2007, The Joint Commission (TJC) started the process referred to as "Ongoing Professional Practice Evaluation" (OPPE).^{12,13} TJC states that credentialing and privileging processes must involve "a series of activities designed to collect, verify, and evaluate data relevant to a provider's professional performance."^{1,12,13} This data must be used for significant decisions, such as "recommendations to grant or deny initial and renewed privileges or to limit and revoke a provider's privileges."^{1,12,13} OPPE was one of the major mechanisms created by TJC to define the data needs.^{1,12,13} OPPE required that practitioner-specific performance data be collected.^{12,13} Hospitals systems and radiology departments scrambled to identify existing data that could be used to meet TJC requirements for OPPE.^{14,15} The error rates available for individual radiology practitioners calculated by RADPEER or RADPEER-like systems seemed to be a perfect mechanism for departments and health care systems to meet the requirements in radiology. This led to the unfortunate mixing together of the practice of error-review for both quality improvement as well as professional practice evaluation. Some publications even advocated a "more penetrating faculty surveillance" in which score-based peer review data were used for counseling, remediation, sanctions, restrictive privileges, and termination.¹⁶ The article highlighted that several practicing radiologists had been terminated based on data from score-based peer review data.¹⁶

Another contributing factor to the wide-spread growth of score-based peer review systems was that the ACR began to require that departments practice score-based peer review as a condition for successful application for ACR modality accreditation—such as for magnetic resonance imaging (MRI), nuclear medicine and positron emission tomography (PET), computed tomography (CT), ultrasound, and others.³ Therefore, to achieve ACR modality accreditation, departments had to implement peer-based peer review.

Discontent With Score-Based Peer Review

The original advocates for score-based peer review never claimed that it was an important improvement tool.^{10,11} They advocated that it was a way to practice peer review with minimal work added and that it was a way to meet regulatory requirements.^{10,11} However, over the past 15 years, there has been growing documentation of problems with the way that score-based peer review was being practiced in radiology across the USA (Table 1).^{1-9,12-29} The confounding factor of individual provider error rates being calculated from score-based peer review and used for OPPE to potentially identify outlier performance also turned score-based peer review into something not originally intended by its creators.

Low Error Rates/Under-Reporting

One of the fundamental problems with calculating error rates and then potentially using them to judge individuals pertaining to provider competence was that radiologists stopped reporting or under-reported errors.¹⁻⁸ It has been shown in multiple studies that errors occur in approximately 5%-20% of radiology reports.^{3,17,18} However, when you looked at the number of cases rated a "4" or "misinterpretation" based on national RADPEER data, the error rates were magnitudes smaller. You were as likely to get struck by lightning as you were to see a major (level 4) error reported via score-based peer learning. This is understandable, given that radiologists knew that they might be putting their friends and colleagues' careers at risk by reporting errors.

Another confounding factor leading to under reporting was that score-based peer review relied on the random auditing of cases.^{10,11} The argument was that for the error rates to be "representative" or "valid," the rate needed to be calculated by random audit.^{2,19,20} However, it has been shown that the yield for cases demonstrating significant

Table 1 Issues Identified With Score-Based Peer Review

- Low Error Rates/under-reporting
- Erosion of Trust
- Radiologist's Negative Perception of Score-based Peer Review
- Lack of Connection of Score-Based Peer Review to Learning and Improvement
- Inaccuracy of Score-Based Peer Review

errors is extremely low for random audits.^{2,4} Cases with significant system issues or errors are much more likely to be identified via other mechanisms—such as identification during interdisciplinary conferences, consultation with referring physicians, when viewing comparisons on follow-up studies, complaints to radiology leadership, pathology-radiology correlation, or incident reporting systems.^{2,4} One study showed that zero out of 1690 randomly audited cases yielded a case with a significant error.²¹ Another showed a positive yield 3.88% from the random review of 11,030 cases and most of that 3.88% were related minor issues with report wording.⁴

One of the most ironic aspects of this under-reporting is that the original premise for which error rates from score-based peer review were to be used OPPE was the ability to identify outlier low performing providers. Because errors were not being reported, outlier performers were almost never identified via OPPE mechanisms. In the relatively rare event of poor clinical performance, those individuals were typically identified via other mechanisms and yet appeared as performing within normal standards by data from score-based peer review.

Erosion of Trust

Quality improvement systems are based on trust.¹⁻⁸ For individuals to share their concerns about the system or individual cases and to actively report issues, they must trust the system.¹⁻⁸ The erosion of trust leads to lack of transparency and lack of engagement to help make the system better. The lack of trust resulted in the system being viewed as inherently punitive, pitting colleagues against one another, and distracting from the goal of improving care.³ Discretionary effort to help improve the system also erodes.

Radiologist's Negative Perception of Score-Based Peer Review

Multiple publications highlighted survey data conducted to determine radiologists' perception of score-based peer review. Multiple publications showed that score-based peer review was perceived as punitive.^{4,22} Score-based peer review was viewed as "a waste of time" and only being performed to meet requirements.^{23,24} Another study showed that 45% of US radiologists reported being dissatisfied with peer review because insufficient learning and inaccurate representation of performance.²⁵

Lack of Connection of Score-Based Peer Review to Learning and Improvement

One of the major complaints about score-based peer review was the lack of any activity aimed at or leading toward quality improvement.³ Multiple publications showed lack of evidence as well as negative impression regarding score-based peer review leading to any sort of improvement.^{3,21,22,25,26} There was no impact on patient outcomes.²⁷

Inaccuracy of Score-Based Peer Review

There were also fundamental challenges with score-based peer review, including difficult removing bias from ratings,²⁸ and questions about the use of expert consensus as the "gold standard" for assessing interpretation accuracy.¹⁰ Peer review was also shown not to be inherently consistent.^{3,17,18}

Movement to Peer Learning

The recognition of these multiple problems was also associated with a ground roots push for change to a new system. That new system was eventually called peer learning. The number of departments switching to peer learning started small and built over time. One of the key events that helped propel peer learning to highly penetrate the radiology community was the 2015 Institute of Medicine's publication entitled "Improving Diagnosis in HealthCare."²⁹ In this article, the authors stated that "health care organizations should adopt policies and practices that promote a non-punitive culture that values open discussion and feedback of diagnostic performance."²⁹ The publication focused on the underappreciated problem of diagnostic error in health care⁵ and called out the ineffectiveness of traditional quality assurance processes that focus on evaluating medical error with an emphasis on individual error.⁵ The article was a rally cry supporting the movement to peer learning and called out the flaws of score-based peer review approach to diagnostic error.⁵

A number of articles pointed out the important components that radiology practices must consider with when switching from score-based peer review to peer learning.²⁻⁴ Those are summarized below and listed in Table 2.

Sequestering learning and improvement activities from evaluation for individual provider competence. Deliberate efforts need to be made to create an event or case-based review of potential learning opportunities for the purpose of learning and improvement that is and appears to be separate from those professional practice activities that evaluate individual provider competence, behavior issues, or defined requirements, such as OPPE.¹⁻⁴ This pertains not only for radiology practices but for entire medical staff

Table 2 Key Issues When Transitioning From Peer Review to Peer Learning

- Sequestering learning and improvement activities from evaluation for individual provider competence
- Moving from random review of radiology cases to actively inclusion of identified learning opportunities
- Abandon numerical scoring of errors
- Define processes for feedback and learning
- Defined structure and governance
- Linking the peer learning process with performance improvement

operations.¹ These changes are necessary for hospital systems to become learning organizations.

Given the history of links between score-based peer review programs and professional practice evaluation activities, such as OPPE, it is particularly important that error rates or any other performance parameters about an individual provider based on peer learning data not be provided to the medical staff office for the purposes of OPPE.² In lieu of error rates for the purposes of OPPE, many practices who conduct peer learning programs will share data on program participation with established targets. Potential metrics and targets may include attendance or attestation of review of video recordings or materials generated from peer learning conferences, contribution of cases to peer learning conferences, group goals related to number of cases submitted or reviewed for peer learning, or willingness to have their cases reviewed as part of the peer learning process.²⁻⁴

Moving from random review of radiology cases to actively inclusion of identified learning opportunities. As previously described, the lack of utility of randomized review of radiology cases to produce learning opportunities is well documented.²⁻⁴ The Royal College of Radiologists in the United Kingdom abandoned score-based peer review throughout the National Health Service. Random review of cases has largely been replaced over time with actively identified cases with learning opportunities as a mechanism for peer learning in the US as well.

Abandon numerical scoring of errors. The numerical classification of errors is at the core of score-based peer review. In the descriptions of RADPEER, processes and governance for how the original interpreting radiologists can appeal a reviewer score were emphasized and detailed.^{2,5-8,11,16,19,20,23,24,30-32} There were also multiple interventions to update and improve the scoring system.^{2,5-8,11,16,19,20,23,24,30-32} However, the use of a numerical system created defensiveness and eroded trust.^{2,4} In addition, it was subjective and unreliable and gave a false impression of accuracy.^{2,5,6,17,18,33-37} Discussion at peer review committee meetings and learning conferences were often dominated by arguments about what score was given and whether it was appropriate. This often distracted from the more important discussions pertaining to the nature of the error and how to prevent it from occurring again in the future.²

Most of those radiology practices conducting peer learning have abandoned numerical scoring systems. Many have now replaced them with descriptive categories that facilitate learning. One organization saw a marked increase in actively submitted cases when they switched from the terminology of “error” to “learning opportunity.”⁴ One of the proposed categories to sort learning opportunities used the following categories: perception, interpretation, technical factors, reporting, communication, radiologist recommendations, or other process-related factors. Most organizations also have a classification for “great calls.”²⁻⁴ Great calls are defined as cases in which a radiologist has made a correct finding and interpretation but there is a

reasonable chance that another radiologists would likely not have done so.^{2,5} There is often very little difference in the nature of cases that are learning opportunities and those that are great calls, with the exception of the radiologists identifying or not identifying the pertinent findings. The opportunity for group learning is the same.

Define processes for feedback and learning. A key component of a peer learning program is establishing and defining the mechanisms by which the practice can learn from the identified cases.

One of the key components of the learning program is establishing a process by which the original reading radiologist receives confidential feedback about the submitted case.^{3,4} For individuals to learn from their errors, they must be aware of them.^{3,4} Some peer learning programs have an electronic system that notify the reading radiologist immediately when a case has been submitted, containing the description of the identified issues.^{3,4} Other programs rely on notification from the person in charge of the peer learning program.²⁻⁴ Regardless of the process, the delivery of the feedback should be done in a delicate and professional manner, as receiving such feedback can be anxiety producing for the recipient even when the emphasis is on learning and improvement.²⁻⁴

In addition to a process for individual feedback, the other core component of a peer learning system is establishing a process for group learning. This has historically and most commonly been achieved via a peer learning conference. In peer learning conferences, cases are shared to drive learning and improvement.^{2,21-24,30}

The way that the peer learning conferences or other learning process are organized often depends upon the size and degree of sub-specialization of the practice. In small departments that practice general radiology, a single peer learning conference may be most beneficial. In larger academic departments with multiple divisions for different organ systems and modalities, separate conferences for each subspecialty or a rotating schedule of subspecialty topics for the entire department maybe more beneficial.² Often the root causes of both errors and identified system issues pertain to all subspecialties even if the imaging indication is subspecialty related. So, often all department members can benefit from a conference even if they do not practice that specific area of subspecialty.²

With increasing frequency of radiology practices having multiple physical locations, shift-work, and remote workers, it has become more difficult for all radiologists to attend a single physical conference. These issues were magnified by the COVID-19 pandemic and changes in post-pandemic radiology practice. These challenges have spawned several solutions. Some groups conduct a peer learning conference in person but also either video-tape the conference for future viewing; conduct conferences virtually, broadcast an in person conference via video conferencing links; or send out a prepared document summarizing the cases discussed for those who are unable to attend to review at a future time.²⁻⁴ Other departments

have created electronic content to be viewed by practicing radiologists asynchronously, when they have time, in lieu of a conference.

Peer learning activities often become one of the most popular and well attended sessions in the department. In addition to practicing radiologists, other attendees often include trainees, such as residents, fellows, and rotating students. Exposing radiology trainees to peer learning is an important aspect of their preparation for clinical practice.

Cases for peer learning conferences should be presented in a de-identified fashion. The de-identification process applies both to the patient identifying information as well as identifying information about the interpreting radiologist. The focus of the discussion is always the learning opportunity, group learning, and preventing the identified issue from occurring again. It is never a discussion of the performance of an individual practitioner.

Peer learning conferences are not the same as interesting case conferences. The focus of interesting case conferences is the discussion of unusual diagnosis or unusual presentations of more common cases. Ideal cases for peer learning conferences are those in which the imaging findings are easily missed or easily misinterpreted (Fig. 1). Important cases to be discussed at peer learning conferences also include those in which system-issues are identified. Such cases may demonstrate issues with case scheduling, protocoling (Fig. 2), image acquisition, delay in interpretation, lack of effective communication of the results (Fig. 3), patient experience, or equity.

Defined structure and governance. Peer learning programs require time and organization. There needs to be a peer learning leader, possibly supported by a peer learning committee.^{2,3} In larger departments, there are often peer learning leaders for each division. These leaders are responsible for overseeing the processes for case collection, organization of the materials for peer learning conferences, overseeing the process for providing individual feedback, and other learning processes.

Linking the peer learning process with quality improvement. When system-related issues are identified through the peer learning process, it is important that there is a formal link between the practice's peer learning program and the processes for quality improvement.²⁻⁴ This link may occur when the case is identified, reviewed, or discussed at the peer learning conference. Practices should have a defined process for problem solving accountability to address the identified issues. An action plan should be created with an action plan owner, timeline with a targeted completion date, and process for follow-up should be established.

Regulatory Myths

There are several common questions encountered when talking with physicians who are contemplating switching to

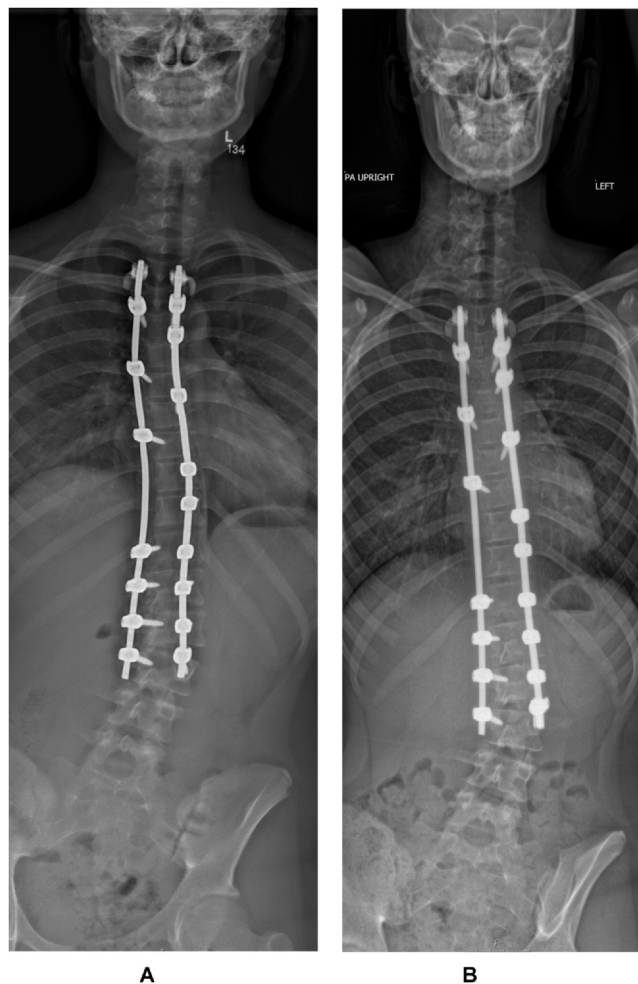


Figure 1 Great Call. Frontal scoliosis radiograph (A) shows what appears to be bent posterior spinal rods, as compared to the straight appearance of the rods on previous study (B). This was submitted as a “Great Call” as radiologists correctly identified that this was artifact secondary to motion during image acquisition on an EOS system and not a true interval bend in the rods.

or starting a peer learning system. One is “How does one convince the medical staff office to accept a peer learning process and not demand numerical “accuracy” data from peer review for OPPE?” A related question is often “How do you switch to peer learning and still meet The Joint Commission’s (TJC) requirements for peer review?” The underlying factors related to these questions is more based on myth than actual facts.

In reviewing the over 1500 TJC elements for which health care organization can be cited for deficiencies, there is only one reference to peer review. It is in the section which discusses the requirements for OPPE.^{12,13,38} The statement is that peer review or peer recommendation **can** be used for the purposes of OPPE.^{12,13,38} It is not stated that peer review data have to be used for OPPE. It does not state that peer review has to be performed at all. And, it in no way states how one must conduct peer review. In summary, there are no TJC requirements that organizations must conduct peer review or that peer review data must be used

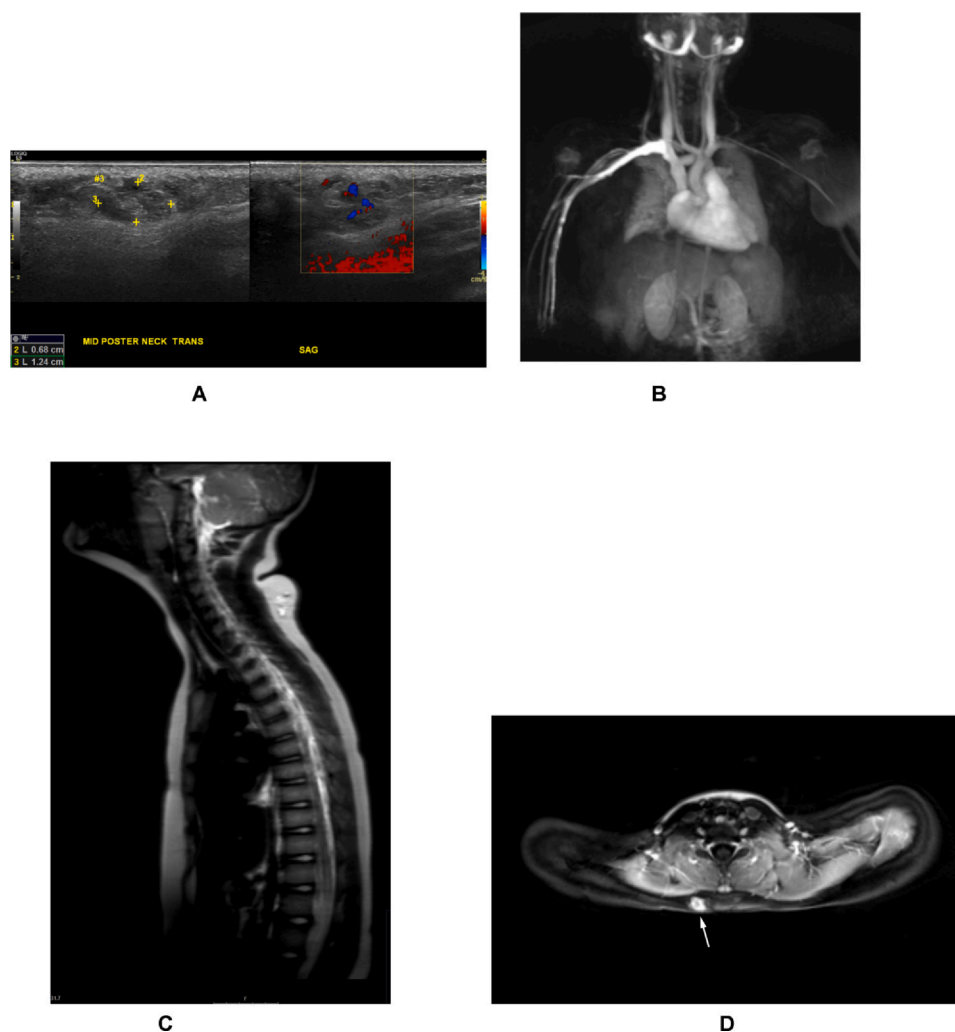


Figure 2 System issue - suboptimal MRI protocolling. (A) Ultrasound identified soft tissue nodules (cursors) with increased venous flow in the subcutaneous tissues in the posterior aspect of the cervical thoracic junction. MRI was ordered to further evaluate. MRI was ordered, protocolled, and performed as “MRI chest with contrast and MRA.” (B) High-quality image from MRA of the chest. (C) Sagittal T2-weighted HASTE and axial T1-weighted post contrast image (D) shows lesion (arrows) in subcutaneous tissues at edge of field of view. Optimal protocol would have focused on region of interest with surface coil and smaller field of view, rather than routine chest protocol with field of view including the entire chest. Quality improvement project arose out of review of this and other examples to redesign process for clinician ordering as well as protocolling processes. HASTE, half fourier single shot turbo spin-echo; MRA, magnetic resonance arteriogram; MRI, magnetic resonance imaging.

for OPPE. There is also no language that prohibits organizations from using a system of peer learning.

Similarly, there are no requirements in the Conditions for Participation defined by the Center for Medicare and Medicaid Services that state that peer review is required or would prohibit a system from practicing peer learning over peer review.^{1,39} There are no requirements of the American Board of Radiology (ABR) that radiologists or radiology practices perform peer review or would prohibit the practice of peer learning. Peer review or peer learning can be an activity that counts toward meeting requirements for *Maintenance of Certification Part 4: Practice Quality Improvement*.⁴⁰ Otherwise, there is no mention of peer review in materials of the ABR.

In addition, the ACR has now embraced peer learning and has an ACR Peer Learning Committee. The previous

requirements for ACR Modality Accreditation that required score-based peer review have been revoked.³ Applicants for ACR Modality Accreditation can now practice either peer learning or score-based peer review.³

In summary, there are no overarching regulatory requirements by TJC, Center for Medicare and Medicaid Services, ABR, or ACR that would prohibit a radiology practice from practicing peer learning.

Peer Learning - The Current State

Over the past 15 years, there has been a steady progression of the development and adoption of the peer learning approach to event-based learning and improvement. In

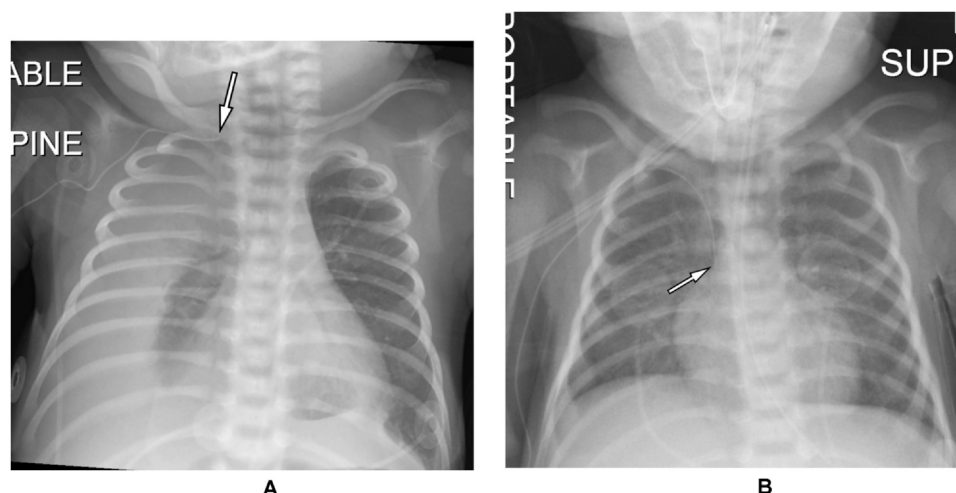


Figure 3 Communication issue - hemothorax secondary venous rupture from PICC line. Chest radiograph (A) with previous comparison (B) show development of new pleural effusion and change in position of right upper extremity PICC line with new laxity in line and tip (arrows) having moved more superiorly, now positioned over the right apex. Both findings were accurately described by the reading radiologist. Discussion of case revolved around radiologist raising possibility of hemothorax and nuances concerning communication of the information. PICC, peripherally inserted central catheter.

January 2020, the ACR sponsored a Peer Learning Summit that both reviewed the progress made as well as further mapped out the path forward.³ The ACR has also created a Peer Learning Committee, created minimal requirements for peer learning,⁴¹ and sponsored many peer learning webinars to help practices.

According to a recent survey of ACR members, 53% are now practicing peer learning.⁴² Of those practicing peer learning, the majority view the practice favorably: practice supports an improved culture of safety and wellness (89%), fosters continuous improvement (86%), helps identify learning opportunities from routine practice (83%), and has a high net promoter score.⁴²

The American College of Radiology's Minimal Requirements for Peer Learning

To help guide practices on meeting ACR requirements for practicing peer learning as it pertains to receiving credit for ACR Modality Accreditation, the ACR recently published the *ACR Minimal Requirements for Peer Learning*.⁴¹

The policy is set up like many regulatory policies—whereby you create the details of your policy and then document that you follow the policy and met the self-established goals. The policy is summarized below and in Table 3.

Having a written peer learning practice policy is required.⁴¹ That policy must include statements pertaining to the culture (emphasize learning and minimize blame) and goals (improvement of services).⁴¹ There must be definition of peer learning opportunities (discrepancies, great calls).⁴¹ The mechanism of case identification—such as cases identified via routine work, case conferences, incident reports—should be defined.⁴¹ Cases should not be identified via

random review only.⁴¹ A description of the program structure and organization should be included.⁴¹ This should include definition of leadership roles (who is in charge of the peer learning program) and what the responsibilities and

Table 3 Summary of the ACR Minimum Requirements for Peer Learning Practice Policy*

- **Written Practice Policy** required with definitions of:
 - Culture (emphasize learning and minimize blame) and goals (improvement of services)
 - Peer learning opportunities (discrepancies, great calls)
 - Mechanism of case identification (not randomly selected)
 - Program Structure & Organization (roles, responsibilities, workflow)
 - Targets (expectations for minimal radiologist participation – case submission or learning activities)
 - Link to Quality Improvement (how do system issues identified via peer learning get translated into corrective action plans)
 - Position on Data Reporting (sequestration of peer learning from evaluation of individual practitioner's performance evaluation)
- **Annual Report of Peer Learning Activities** which includes:
 - Number of cases submitted for peer learning
 - Number/percentage of radiologist meeting established targets
 - Statement as to whether Practice Policy minimal standards were or were not met
 - Quality Improvement Action Plans & Accomplishments

* Adopted from.⁴¹

support for those leaders are.⁴¹ The workflow for peer learning case submission, communication to the interpreting radiologist, and process for peer learning activities should be defined.⁴¹ The policy should also define the minimal targets for radiologist participation, such as defining any requirements for number of cases submitted or participation in a defined percentage of learning activities.⁴¹ The number and frequency of peer learning activities, such as learning conferences, should also be defined.⁴¹ The policy should define the process for coordination with the quality improvement infrastructure of the practice.⁴¹ The mechanism by which system issues identified via peer learning are translated into quality improvement action plans should be defined. The policy also requires a statement on sharing of data from peer learning. Specifically, the policy should define that data from peer learning is sequestered from and not shared with processes related to individual practitioner's performance evaluation, such as OPPE.⁴¹

In addition to the requirement for a written practice policy, the other component of the ACR minimal standards for peer learning is the requirement for an annual documentation of program activities.⁴¹ This should include a description of the total number of cases submitted; the number and percentage of radiologists meeting the defined targets; a statement as to whether the practice did or did not meet the defined minimal standards; and a summary of the quality improvement efforts and accomplishments occurring that year.⁴¹

Future Opportunities and Challenges

The practice of peer learning is still developing and evolving. There are many challenges innate to the transition of the radiology community from a system of score-based peer review to one of peer learning. There will be required cultural transformations for the community to move from processes perceived as punitive to those focused on learning and improvement. It will take time for trust to be completely developed. The support and understanding of medical staff offices and regulatory agencies will both be key and evolving.

One of the challenges historically both with a peer review or peer learning approach has been protection of the identified information about an error from discovery and potential for malpractice claims. Most US states have statutes that protect information obtained for quality improvement purposes from discovery or introduction as evidence in malpractice cases. It is important for peer learning programs to be designed to meet the criteria for such protections based on the laws and statutes of the particular state or states in which the group is practicing.

One additional future opportunity is further partnership with information technology vendors in the radiology space.³ Further development of informatic programs that allow peer learning to be practiced seamlessly and efficiently during clinical work will be incredibly helpful.

The conceptual premise of peer learning is that the circumstances that lead to an error by a single radiologist in a practice are apt to happen to other practicing radiologists as well. By having shared learning from these cases within the group, future errors can be avoided and system issues corrected. By this same logic, it is likely that radiologists in different practices around the country and, in fact, around the world are highly likely to make the same types of errors or encounter similar system challenges. There is a great potential to improve care by sharing peer learnings on a national or international basis. ACR is sponsoring webinars where peer learning cases in a given radiology subspecialty—such as pediatric radiology or neuroradiology—are presented by multiple institutions to share learnings more broadly as well as demonstrate how peer learning conferences can be executed. There is also opportunity for teaching case databases populated with peer learning-type cases. Alternatively, radiology newsletters highlighting peer learning cases might be helpful. The Royal College of Radiologists has such a newsletter shared throughout the United Kingdom.⁴³

References

1. Sandborg CI, Hartman GE, Su F, et al: Optimizing professional practice evaluation to enable a nonpunitive learning health system approach to peer review. *Pedia Qual Saf* 6:e375, 2020
2. Donnelly LF, Larson DB, Heller III RE, et al: Practical suggestions on how to move from peer review to peer learning. *AJR* 210:578-582, 2018
3. Larson DB, Broder JC, Bhargavan-Chatfield M, et al: Transition from peer review to peer learning: report of the 2020 peer learning summit. *J Am Coll Radiol* 11:1499-1508, 2020
4. Donnelly LF, Dorfman SR, Jones J, et al: Transition from peer review to peer learning: experience in a radiology department. *J Am Coll Radiol* 15:1143-1149, 2018
5. Larson DB, Donnelly LF, Podberesky DJ, et al: Peer feedback, learning, and improvement: answering the call of the Institute of Medicine's report on diagnostic error. *Radiology* 283:231-241, 2017
6. Larson DB, Nance JJ: Rethinking peer review: what aviation can teach radiology about performance improvement. *Radiology* 259:626-632, 2011
7. Kruskal JB, Eisenberg RL, Brook O, et al: Transitioning from peer review to peer learning for abdominal radiologists. *Abdom Radiol* 41:416-428, 2016
8. Halsted MJ: Radiology peer review as an opportunity to reduce errors and improve patient care. *J Am Coll Radiol* 1:984-987, 2004
9. Institute of Medicine, Committee on Quality of Health Care in America. *To Err Is Human: Building a Safer Health System*. Washington, DC: National Academies Press, 2000
10. Borgstede JP, Lewis RS, Bhargavan M, et al: RADPEER quality assurance program: a multifacility study of interpretive disagreement rates. *J Am Coll Radiol* 1:59-65, 2004
11. Jackson VP, Cushing T, Abujudeh HH, et al: RADPEER™ scoring white paper. *J Am Coll Radiol* 6:21-25, 2009
12. Joint Commission. MS.08.01.03: Ongoing professional practice evaluation information is factored into the decision to maintain existing privileges(s), to revise existing privileges(s), or to revoke an existing privilege prior to or at the time of renewal. Oakbrook Terrace, IL, The Joint Commission, 2019.
13. Joint Commission. MS.06.01.01: Prior to granting a privilege, the resources necessary to support the requested privilege are determined to be currently available, or available within a specific time frame. Oakbrook Terrace, IL, The Joint Commission, 2019.

14. Donnelly LF, Strife JL: Performance-based assessment of radiology faculty: a practical plan to promote improvement and meet JCAHO standards. *AJR* 184:1398-1401, 2005
15. Donnelly LF: Performance-based assessment of radiology providers: promoting improvement in accordance with the 2007 Joint Commission standards. *J Am Coll Radiol* 4:699-703, 2007
16. Hussain S, Hussain JS, Karam A, et al: Focused peer review: the end game of peer review. *J Am Coll Radiol* 9:430-433, 2012
17. Smith JT: It's not about the errors, it's about the learning: How the Royal College of Radiologists has developed a Radiology Events and Learning process in the United Kingdom. *J Med Imaging Radiat Oncol* 66:185-192, 2022
18. The Royal College of Radiologists. Quality assurance in radiology reporting: peer feedback. London: The Royal College of Radiologists, 2014
19. Fotenos A, Nagy P: What are your goals for peer review? A framework for understanding differing methods. *J Am Coll Radiol* 9:929-930, 2012
20. Larson PA, Pyatt RS, Grimes CK, et al: Getting the most out of RADPEER™. *J Am Coll Radiol* 8:543-548, 2011
21. Itri JN, Donithan A, Patel SH: Random versus nonrandom peer review: a case for more meaningful peer review. *J Am Coll Radiol* 15:1045-1052, 2018
22. Johnson CD, Krecke KN, Miranda R, et al: Quality initiatives developing a radiology quality and safety program: a primer. *Radiographics* 29:951-959, 2009
23. Eisenberg RL, Cunningham ML, Siewert B: Survey of faculty perceptions regarding a peer review system. *J Am Coll Radiol* 11:397-401, 2014
24. Abujudeh H, Pyatt RS, Bruno MA, et al: Radpeer peer review: relevance, use, concerns, challenges, and direction forward. *J Am Coll Radiol* 11:899-904, 2014
25. Lee CS, Neumann C, Jha P, et al: Current status and future wish list of peer review: a national questionnaire of US radiologists. *AJR* 214:493-497, 2020
26. Trinh TW, Shinagare AB, Khorasani R: Yield of learning opportunities from a radiology random peer review program. *AJR* 211:630-634, 2018
27. Sheu R, Feder E, Balsin I, et al: Optimizing radiology peer review: a mathematical model for selecting future cases based on prior errors. *J Am Coll Radiol* 7:431-438, 2010
28. Cascade PN, Kazerooni EA, Gross BH, et al: Evaluation of competence in the interpretation of chest radiographs. *Acad Radiol* 8:315-321, 2001
29. Balogh EP, Miller BT, Bll JR, eds. Board on Health Care Services, Institute of Medicine: Improving Diagnosis in Health Care. Washington, DC, The National Academy of Sciences, The National Academies Press, 2015.
30. Alkasab TK, Harvey HB, Gowda V, et al: Consensus-oriented group peer review: a new process to review radiologist work output. *J. Am. Coll. Radio.* 11(2):131-138, 2014
31. Mahgerefteh S, Kruskal JB, Yam CS, et al: Peer review in diagnostic radiology: current state and a vision for the future 1. *Radiographics* 5:1221-1231, 2009
32. Butler GJ, Forghani R: The next level of radiology peer review: enterprise-wide education and improvement. *J Am Coll Radiol* 10:349-353, 2013
33. The Royal College of Radiologists. Standards for learning discrepancies meetings: The Royal College of Radiologists, 2014.
34. Mucci B, Murray H, Downie A, et al: Inter-rater variation in scoring radiological discrepancies. *Br J Radiol* 86:20130245, 2013
35. Bender LC, Linnau KF, Meier EN, et al: Interrater agreement in the evaluation of discrepant imaging findings with the Radpeer system. *Am J Roentgenol* 199:1320-1327, 2012
36. Semelka RC, Ryan AF, Yonkers S, et al: Objective determination of standard of care: use of blind readings by external radiologists. *Am J Roentgenol* 195:429-431, 2010
37. National Advisory Group on the Safety of Patients in England. A promise to learn-a commitment to act. Improving the safety of patients in England. London: NHS England, 2013
38. Joint Commission. MS.06.01.05: The decision to grant or deny a privilege(s) and/or to renew an existing privilege(s), is an objective, evidence-based process. Oakbrook Terrace, IL, The Joint Commission, 2019.
39. Center for Medicare & Medicaid Services (CMS) requirements for hospital medical staff privileging. Baltimore, ML, CMS, 2004.
40. Donnelly LF, Mathews VP, Laszakovitz DJ, et al: Recent changes to ABR Maintenance of certification Part 4 (PQI): Acknowledgement of radiologists' activities to improve quality and safety. *J Am Coll Radiol* 13:184-187, 2016
41. Available at: <https://www.acr.org/-/media/ACR/Files/Peer-Learning-Summit/Requirements-for-PL-program-accreditation.pdf>. Accessed March 7, 2023.
42. Sharpe Jr RE, Tarrant MJ, Brook OR, et al: Current state of peer learning in radiology: a survey of ACR members. *J Am Coll Radiol* 20:699-711, 2023
43. Available at: <https://www.rcr.ac.uk/posts/radiology-events-and-learning-real-newsletter-launch>. Accessed March 7, 2023.